

RISC-V Matrix-Matrix Multiplication

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Code

0.1 RISC-V Assembly Code

Listing 1: matrix_multiply.asm

```
1 # Purpose : perform matrix multiplication C = A x B
2 # Assumption :
3 # - A is size (m x m) and a[i][j] >= 0 for all i,j
4 # - B is size (m x m) and b[i][j] >= 0 for all i,j
5 # - Each elem in C is not larger than 32-bits
6 #
7 # Inputs : [int] [m x m] A
8 #          [int] [m x m] B
9 #
10 # Return : [int] [m x m] C
11 #
12 # NOTE : preserved/saved registers are s0-s11, sp, and ra
13 # NOTE : nonpreserved are a0-a7, t0-t6
14 # NOTE : Ensure the sp is at a high number for RISC-V
15
16
17 # ~~~~~ compile time ~~~~~
18 .equ M, 16          # one dim of the mat
19
20
21 # ~~~~~ data segment (memory) ~~~~~
22 .data
23
24 # ~~ TC : 2x2 identity matrix ~~
25 #A:    .word 1, 2, 3, 4
26 #B:    .word 1, 0, 0, 1
27 #C:    .word -7:4
28 #answer = 1, 2, 3, 4
29
30 # ~~ TC : 2x2 big zero ~~
31 #A:    .word 1, 2, 3, 4
32 #B:    .word 0, 0, 0, 0
33 #C:    .word -7:4
34 #answer = 0, 0, 0, 0
35
36 # ~~ TC : 2x2 stairs ~~
37 #A:    .word 1, 2, 3, 4
38 #B:    .word 5, 6, 7, 8
39 #C:    .word -7:4
40 #answer = 19, 22, 43, 50
41
42 # ~~ TC : 2x2 swap ~~
43 #A:    .word 1, 2, 3, 4
44 #B:    .word 0, 1, 1, 0
45 #C:    .word -7:4
46 #answer = 2, 1, 4, 3
47
48 # ~~ TC : 3x3 identity matrix ~~
49 #A:    .word 1, 2, 3, 4, 5, 6, 7, 8, 9
50 #B:    .word 1, 0, 0, 0, 1, 0, 0, 0, 1
51 #C:    .word -7:9
52 #answer = 1, 2, 3, 4, 5, 6, 7, 8, 9
53
54 # ~~ TC : 3x3 big zero ~~
55 #A:    .word 1, 2, 3, 4, 5, 6, 7, 8, 9
56 #B:    .word 0, 0, 0, 0, 0, 0, 0, 0, 0
57 #C:    .word -7:9
58 #answer = 0, 0, 0, 0, 0, 0, 0, 0, 0
59
60 # ~~ TC : 3x3 stairs ~~
61 #A:    .word 1, 2, 3, 4, 5, 6, 7, 8, 9
62 #B:    .word 10, 11, 12, 13, 14, 15, 16, 17, 18
63 #C:    .word -7:9
64 #answer = 84, 90, 96, 201, 216, 231, 318, 342, 366
65
66 # ~~ TC : 3x3 identity swap ~~
67 #A:    .word 1, 2, 3, 4, 5, 6, 7, 8, 9
68 #B:    .word 0, 0, 1, 0, 1, 0, 1, 0, 0
69 #C:    .word -7:9
70 #answer = 3, 2, 1, 6, 5, 4, 9, 8, 7
71
72 # ~~ TC : dataset3.h ~~
```

```

73 #A:      .word 0, 3, 2, 0, 3, 1, 0, 3, 2
74 #B:      .word 1, 1, 0, 3, 1, 2, 0, 0, 0
75 #C:      .word -7:9
76 #answer = 9, 3, 6, 9, 3, 6, 9, 3, 6
77
78 # ~~ TC : dataset10.h ~~
79 #A:      .word 0, 3, 2, 0, 3, 1, 0, 3, 2, 3  #1
80 #      .word 2, 0, 3, 3, 1, 2, 3, 0, 0, 1  #2
81 #      .word 1, 1, 2, 3, 1, 2, 3, 1, 1, 3  #3
82 #      .word 2, 2, 0, 1, 3, 2, 2, 2, 0, 0  #4
83 #      .word 1, 0, 1, 3, 3, 0, 3, 3, 3, 3  #5
84 #      .word 0, 3, 2, 1, 2, 2, 0, 0, 3, 0  #6
85 #      .word 1, 1, 0, 3, 3, 1, 2, 3, 3, 0  #7
86 #      .word 1, 2, 1, 0, 1, 2, 2, 1, 0, 3  #8
87 #      .word 1, 0, 2, 2, 1, 1, 1, 1, 1, 1  #9
88 #      .word 2, 0, 3, 1, 1, 2, 2, 3, 3, 1  #10
89
90 #B:      .word 1, 1, 0, 3, 1, 2, 0, 0, 0, 0  #1
91 #      .word 0, 2, 1, 2, 3, 0, 0, 3, 3, 2  #2
92 #      .word 2, 1, 2, 3, 3, 0, 2, 2, 1, 1  #3
93 #      .word 2, 2, 0, 2, 2, 1, 2, 3, 2, 2  #4
94 #      .word 3, 3, 2, 2, 1, 1, 1, 1, 2, 1  #5
95 #      .word 2, 2, 3, 3, 3, 0, 0, 3, 2, 3  #6
96 #      .word 2, 3, 1, 2, 1, 1, 2, 2, 0, 1  #7
97 #      .word 0, 3, 2, 1, 1, 1, 2, 0, 1, 2  #8
98 #      .word 2, 0, 2, 1, 3, 3, 2, 3, 2, 0  #9
99 #      .word 3, 1, 3, 3, 2, 0, 1, 0, 1, 1  #10
100
101 #C:      .word -7:100
102
103
104 #answer =      28, 31, 35, 35, 36, 12, 20, 25, 29, 23  #1
105 #      30, 28, 20, 38, 29, 11, 20, 28, 16, 20  #2
106 #      35, 33, 29, 42, 36, 13, 24, 32, 23, 25  #3
107 #      21, 33, 20, 30, 23, 12, 13, 22, 20, 21  #4
108 #      39, 38, 32, 39, 34, 23, 32, 29, 25, 22  #5
109 #      22, 20, 23, 27, 34, 12, 14, 33, 27, 18  #6
110 #      28, 35, 24, 30, 30, 22, 25, 31, 26, 22  #7
111 #      23, 25, 25, 32, 26, 6, 12, 19, 17, 19  #8
112 #      21, 19, 17, 25, 22, 10, 16, 19, 14, 14  #9
113 #      30, 30, 31, 38, 36, 20, 26, 29, 21, 21  #10
114
115 # ~~ TC : dataset16.h ~~
116
117 A:
118      .word 0, 3, 2, 0, 3, 1, 0, 3, 2, 3, 2, 0, 3, 3, 1, 2  #1
119      .word 3, 0, 0, 1, 1, 1, 2, 3, 1, 2, 3, 1, 1, 3, 2, 2  #2
120      .word 0, 1, 3, 2, 2, 2, 0, 0, 1, 0, 1, 3, 3, 0, 3, 3  #3
121      .word 3, 3, 0, 3, 2, 1, 2, 2, 0, 0, 3, 0, 1, 1, 0, 3  #4
122      .word 3, 1, 2, 3, 3, 0, 1, 2, 1, 0, 1, 2, 2, 1, 0, 3  #5
123      .word 1, 0, 2, 2, 1, 1, 1, 1, 1, 1, 2, 0, 3, 1, 1, 2  #6
124      .word 2, 3, 3, 1, 3, 2, 0, 0, 0, 3, 3, 3, 2, 1, 2, 3  #7
125      .word 1, 0, 0, 0, 0, 1, 2, 2, 1, 1, 3, 3, 3, 1, 1, 2  #8
126      .word 3, 1, 3, 3, 2, 3, 2, 1, 2, 3, 0, 2, 2, 1, 1, 0  #9
127      .word 0, 0, 0, 0, 1, 3, 3, 1, 1, 1, 2, 2, 3, 2, 1, 1  #10
128      .word 1, 1, 3, 0, 2, 2, 1, 3, 2, 1, 2, 2, 1, 3, 1, 3  #11
129      .word 1, 3, 2, 3, 1, 2, 1, 3, 2, 2, 0, 1, 0, 0, 1, 2  #12
130      .word 3, 3, 1, 0, 0, 0, 3, 1, 2, 3, 2, 3, 2, 0, 0, 0  #13
131      .word 0, 0, 3, 1, 3, 0, 0, 0, 3, 1, 1, 1, 1, 2, 1, 2  #14
132      .word 3, 2, 0, 0, 2, 2, 3, 0, 3, 0, 0, 3, 0, 3, 1, 3  #15
133      .word 3, 1, 1, 1, 2, 2, 1, 3, 0, 3, 3, 1, 0, 0, 3, 2  #16
134
135
136 B:
137      .word 1, 1, 0, 3, 1, 2, 0, 0, 0, 0, 0, 2, 1, 2, 3, 0  #1
138      .word 0, 3, 3, 2, 2, 1, 2, 3, 3, 0, 2, 2, 1, 1, 2, 2  #2
139      .word 0, 2, 2, 1, 2, 3, 2, 2, 3, 3, 2, 2, 1, 1, 1, 1  #3
140      .word 2, 1, 2, 2, 3, 3, 3, 0, 0, 3, 2, 3, 2, 3, 1, 2  #4
141      .word 1, 1, 2, 2, 0, 1, 0, 3, 2, 1, 1, 1, 2, 0, 1, 2  #5
142      .word 2, 0, 2, 1, 3, 3, 2, 3, 2, 0, 3, 1, 3, 3, 2, 0  #6
143      .word 1, 0, 1, 1, 2, 2, 1, 1, 2, 2, 1, 2, 3, 3, 1, 3  #7
144      .word 2, 2, 2, 3, 3, 1, 0, 2, 1, 0, 0, 0, 1, 1, 2, 0  #8
145      .word 3, 2, 3, 3, 0, 2, 3, 1, 0, 0, 2, 1, 2, 0, 2, 1  #9
146      .word 1, 2, 3, 1, 3, 2, 1, 0, 0, 0, 0, 0, 2, 2, 0, 2  #10
147      .word 1, 2, 0, 3, 2, 2, 0, 0, 3, 2, 1, 1, 3, 0, 2, 0  #11
148      .word 0, 1, 0, 2, 3, 3, 1, 3, 3, 0, 0, 2, 2, 0, 0, 0  #12
149      .word 1, 0, 0, 1, 3, 0, 2, 1, 3, 2, 2, 1, 3, 2, 0, 1  #13
150      .word 2, 2, 3, 2, 1, 1, 1, 1, 3, 0, 1, 3, 2, 2, 3, 1  #14
151      .word 1, 2, 0, 2, 1, 1, 2, 3, 1, 0, 1, 0, 1, 1, 0, 0  #15
152      .word 2, 0, 3, 0, 3, 0, 3, 2, 2, 3, 3, 2, 1, 0, 2, 2  #16
153
154 C:      .word -7:256      # 16^2 = 256
155
156
157 #answer =
158 # 36, 44, 57, 50, 54, 36, 38, 46, 55, 25, 38, 34, 51, 30, 40, 32  #1
159 # 37, 34, 38, 52, 51, 40, 28, 32, 41, 22, 26, 35, 49, 35, 42, 23  #2
160 # 26, 26, 33, 36, 52, 40, 45, 49, 50, 34, 41, 35, 44, 25, 23, 23  #3
161 # 31, 29, 39, 46, 50, 36, 31, 32, 42, 32, 34, 41, 44, 33, 43, 30  #4
162 # 31, 28, 39, 46, 50, 40, 35, 37, 43, 35, 33, 43, 43, 29, 37, 29  #5
163 # 27, 22, 30, 33, 43, 31, 32, 25, 36, 31, 31, 29, 40, 28, 26, 22  #6

```

```
164 # 29, 42, 48, 51, 65, 52, 43, 54, 63, 34, 42, 44, 56, 33, 38, 32 #7
165 # 26, 22, 23, 38, 49, 32, 26, 30, 43, 22, 24, 27, 45, 24, 26, 17 #8
166 # 35, 35, 47, 51, 59, 59, 43, 42, 43, 28, 37, 43, 56, 48, 36, 32 #9
167 # 28, 19, 28, 34, 46, 34, 28, 34, 45, 20, 29, 28, 50, 32, 26, 21 #10
168 # 37, 38, 51, 50, 55, 45, 38, 49, 56, 28, 38, 40, 50, 29, 44, 26 #11
169 # 32, 35, 50, 43, 53, 44, 41, 41, 34, 24, 35, 34, 39, 33, 34, 29 #12
170 # 21, 33, 31, 45, 48, 42, 27, 29, 40, 17, 21, 32, 45, 30, 29, 26 #13
171 # 26, 27, 38, 33, 29, 31, 32, 31, 35, 25, 29, 29, 34, 15, 25, 23 #14
172 # 34, 28, 44, 45, 41, 41, 37, 45, 45, 17, 34, 44, 46, 30, 43, 29 #15
173 # 31, 36, 37, 50, 54, 44, 28, 40, 38, 22, 27, 28, 45, 32, 36, 22 #16
174
175
176 # ~~ TC : dataset50.h ~~
177 #A:
178 # .word 0, 3, 2, 0, 3, 1, 0, 3, 2, 3, 2, 0, 3, 3, 1, 2, 3, 0, 0, 1
179 # .word 1, 1, 2, 3, 1, 2, 3, 1, 1, 3, 2, 2, 0, 1, 3, 2, 2, 2, 0, 0
180 # .word 1, 0, 1, 3, 3, 0, 3, 3, 3, 3
181
182 # .word 0, 3, 2, 1, 2, 2, 0, 0, 3, 0 # 2
183 # .word 1, 1, 0, 3, 3, 1, 2, 3, 3, 0, 1, 2, 1, 0, 1, 2, 2, 1, 0, 3
184 # .word 1, 0, 2, 2, 1, 1, 1, 1, 1, 1, 2, 0, 3, 1, 1, 2, 2, 3, 3, 1
185
186 # .word 3, 2, 0, 0, 0, 3, 3, 3, 2, 1, 2, 3, 1, 0, 0, 0, 0, 1, 2, 2
187 # .word 1, 1, 3, 3, 3, 1, 1, 2, 3, 1, 3, 3, 2, 3, 2, 1, 2, 3, 0, 2
188 # .word 2, 1, 1, 0, 0, 0, 0, 0, 1, 3
189
190 # .word 3, 1, 1, 1, 2, 2, 3, 2, 1, 1 # 4
191 # .word 1, 1, 3, 0, 2, 2, 1, 3, 2, 1, 2, 2, 1, 3, 1, 3, 1, 3, 2, 3
192 # .word 1, 2, 1, 3, 2, 2, 0, 1, 0, 0, 1, 2, 3, 3, 1, 0, 0, 0, 3, 1
193
194 # .word 2, 3, 2, 3, 2, 0, 0, 0, 0, 0, 3, 1, 3, 0, 0, 0, 3, 1, 1, 1
195 # .word 1, 2, 1, 2, 3, 2, 0, 0, 2, 2, 3, 0, 3, 0, 0, 3, 0, 3, 1, 3
196 # .word 3, 1, 1, 1, 2, 2, 1, 3, 0, 3
197
198 # .word 3, 1, 0, 0, 3, 2, 1, 3, 3, 3 # 6 # 6
199 # .word 1, 0, 1, 1, 2, 1, 0, 1, 1, 2, 2, 3, 1, 2, 2, 2, 0, 1, 3, 3
200 # .word 3, 2, 2, 1, 0, 1, 2, 0, 1, 1, 1, 1, 2, 3, 2, 2, 3, 3, 0, 0
201
202 # .word 2, 0, 0, 0, 3, 0, 1, 0, 3, 0, 0, 0, 3, 0, 0, 2, 0, 2, 0, 0
203 # .word 2, 3, 2, 0, 0, 3, 3, 2, 1, 1, 0, 2, 0, 0, 3, 3, 2, 3, 3, 0
204 # .word 1, 0, 2, 2, 0, 3, 3, 1, 1, 0
205
206 # .word 2, 3, 2, 1, 1, 0, 1, 2, 1, 2 # 8
207 # .word 2, 0, 0, 1, 0, 1, 1, 0, 1, 0, 2, 3, 3, 2, 0, 0, 1, 3, 0, 3
208 # .word 3, 0, 0, 0, 0, 3, 3, 1, 0, 0, 3, 3, 2, 1, 2, 1, 3, 3, 0, 1
209
210 # .word 3, 0, 2, 3, 1, 3, 3, 3, 3, 3, 0, 1, 1, 3, 0, 2, 2, 3, 1, 2
211 # .word 2, 2, 1, 3, 3, 0, 3, 0, 0, 2, 0, 2, 3, 0, 1, 3, 2, 2, 0, 0
212 # .word 2, 3, 0, 2, 2, 2, 3, 1, 0, 3
213
214 # .word 3, 3, 3, 1, 0, 3, 3, 2, 0, 3 # 10
215 # .word 2, 0, 3, 0, 2, 0, 0, 2, 2, 1, 0, 2, 1, 1, 1, 1, 2, 3, 3
216 # .word 3, 0, 0, 3, 3, 3, 2, 3, 3, 1, 2, 2, 3, 1, 2, 1, 1, 3, 0, 1
217
218 # .word 2, 0, 2, 0, 0, 1, 3, 2, 0, 1, 3, 2, 3, 3, 0, 0, 0, 1, 0, 3
219 # .word 3, 2, 2, 2, 1, 1, 2, 2, 1, 3, 2, 0, 1, 3, 2, 0, 2, 1, 3, 0
220 # .word 0, 0, 1, 3, 3, 2, 2, 2, 3, 1
221
222 # .word 0, 0, 1, 1, 2, 1, 3, 1, 1, 2 # 12
223 # .word 2, 3, 2, 3, 0, 2, 3, 3, 0, 3, 0, 0, 1, 0, 0, 0, 1, 3, 1, 1
224 # .word 2, 3, 2, 1, 1, 2, 2, 2, 3, 0, 1, 1, 2, 1, 2, 0, 2, 3, 1, 3
225
226 # .word 0, 1, 1, 3, 0, 2, 3, 0, 1, 2, 3, 2, 0, 0, 3, 3, 2, 1, 1, 2
227 # .word 3, 0, 1, 1, 1, 1, 2, 0, 1, 2, 0, 1, 1, 1, 0, 1, 3, 2, 3, 1
228 # .word 0, 2, 1, 2, 1, 3, 3, 1, 0, 2
229
230 # .word 2, 3, 1, 3, 1, 3, 0, 1, 0, 3 # 14
231 # .word 0, 3, 2, 0, 3, 3, 3, 0, 3, 2, 2, 2, 1, 3, 0, 0, 1, 1, 3, 0
232 # .word 1, 2, 1, 0, 0, 0, 3, 2, 2, 0, 0, 2, 1, 3, 0, 0, 3, 0, 0, 2
233
234 # .word 1, 1, 2, 2, 1, 3, 2, 2, 1, 1, 2, 1, 3, 2, 1, 1, 3, 0, 1, 3
235 # .word 3, 2, 2, 1, 0, 3, 2, 2, 2, 3, 0, 1, 3, 3, 2, 3, 0, 3, 2, 3
236 # .word 1, 1, 0, 0, 0, 2, 3, 0, 3, 0
237
238 # .word 1, 1, 3, 1, 3, 2, 1, 1, 2, 1 # 16
239 # .word 3, 2, 0, 2, 1, 0, 2, 3, 2, 3, 2, 1, 2, 3, 0, 0, 1, 1, 0, 0
240 # .word 2, 1, 0, 1, 2, 2, 2, 2, 0, 3, 3, 1, 0, 0, 0, 0, 3, 1, 1, 0
241
242 # .word 0, 0, 0, 1, 2, 2, 1, 3, 0, 2, 3, 2, 3, 2, 2, 1, 2, 2, 3, 3
243 # .word 1, 3, 0, 2, 2, 3, 3, 1, 2, 2, 2, 3, 1, 1, 1, 0, 0, 0, 3, 0
244 # .word 1, 0, 3, 1, 1, 3, 0, 1, 2, 2
245
246 # .word 0, 0, 3, 3, 3, 3, 2, 1, 0, 0 # 18
247 # .word 1, 0, 2, 0, 1, 1, 0, 0, 3, 3, 2, 1, 1, 1, 0, 1, 1, 2, 2, 1
248 # .word 1, 2, 0, 3, 1, 3, 1, 0, 3, 0, 3, 1, 1, 1, 0, 2, 0, 3, 1, 0
249
250 # .word 1, 0, 2, 0, 2, 3, 3, 3, 1, 2, 3, 2, 2, 0, 1, 1, 0, 3, 3, 1
251 # .word 3, 3, 2, 0, 2, 0, 2, 2, 3, 3, 3, 0, 2, 3, 3, 1, 3, 2, 2, 2
252 # .word 0, 2, 3, 0, 2, 0, 3, 2, 2, 1
253
254 # .word 1, 0, 2, 2, 2, 0, 2, 2, 0, 1 # 20
```

255	#	.word 3,	2,	1,	3,	2,	2,	0,	3,	3,	1,	2,	2,	0,	0,	3,	2,	1,	2,	2,	1
256	#	.word 3,	1,	2,	0,	0,	1,	1,	2,	1,	3,	2,	2,	3,	0,	2,	1,	3,	2,	1,	3
257																					
258	#	.word 2,	3,	3,	1,	2,	1,	2,	2,	0,	0,	0,	3,	0,	2,	3,	1,	0,	0,	2,	3
259	#	.word 3,	2,	2,	1,	3,	3,	2,	2,	2,	1,	1,	3,	3,	3,	0,	1,	1,	2,	1,	1
260	#	.word 1,	0,	2,	1,	2,	0,	2,	1,	2,	3										
261																					
262	#	.word 1,	1,	0,	0,	3,	0,	2,	0,	2,	3					# 22					
263	#	.word 1,	1,	0,	3,	1,	0,	3,	1,	3,	1,	2,	0,	1,	2,	3,	2,	3,	0,	2,	1
264	#	.word 0,	1,	1,	3,	0,	0,	1,	0,	2,	1,	1,	0,	2,	2,	3,	0,	2,	0,	1,	2
265																					
266	#	.word 0,	1,	3,	0,	0,	3,	1,	3,	0,	0,	3,	1,	0,	2,	2,	3,	3,	2,	0,	2
267	#	.word 0,	3,	1,	2,	0,	0,	0,	2,	1,	1,	1,	2,	2,	3,	3,	3,	0,	3,	0,	3
268	#	.word 3,	1,	3,	3,	0,	3,	3,	2,	2,	3										
269																					
270	#	.word 0,	3,	0,	2,	3,	3,	3,	1,	1,	3					# 24					
271	#	.word 1,	0,	3,	3,	2,	0,	1,	3,	2,	0,	1,	1,	3,	2,	1,	3,	1,	2,	0,	0
272	#	.word 1,	1,	0,	0,	0,	1,	1,	3,	1,	2,	2,	2,	3,	1,	1,	0,	0,	2,	3,	3
273																					
274	#	.word 2,	1,	3,	0,	0,	3,	0,	2,	3,	1,	0,	0,	0,	1,	3,	3,	3,	0,	3,	3
275	#	.word 2,	1,	0,	1,	1,	2,	3,	0,	2,	1,	3,	1,	1,	2,	2,	2,	1,	3,	2,	0
276	#	.word 2,	3,	2,	1,	2,	3,	0,	3,	1,	3										
277																					
278	#	.word 2,	3,	2,	2,	3,	2,	3,	3,	0,	1					# 26					
279	#	.word 0,	3,	2,	1,	1,	3,	2,	0,	1,	1,	3,	3,	2,	3,	2,	2,	1,	1,	1,	3
280	#	.word 3,	3,	2,	1,	1,	0,	1,	2,	3,	3,	1,	3,	1,	0,	0,	3,	3,	2,	3,	1
281																					
282	#	.word 3,	1,	0,	1,	3,	1,	0,	0,	0,	2,	0,	2,	1,	3,	2,	3,	2,	1,	0,	1
283	#	.word 3,	2,	0,	0,	2,	3,	0,	2,	0,	1,	0,	1,	2,	1,	3,	2,	2,	1,	3,	3
284	#	.word 2,	1,	0,	1,	0,	0,	0,	2,	1,	2										
285																					
286	#	.word 2,	0,	0,	0,	2,	3,	1,	2,	1,	2					# 28					
287	#	.word 1,	3,	2,	2,	3,	2,	3,	1,	0,	0,	2,	1,	2,	3,	1,	3,	2,	0,	0,	1
288	#	.word 2,	1,	0,	3,	2,	2,	0,	1,	2,	3,	2,	2,	2,	0,	0,	0,	3,	1,	1,	3
289																					
290	#	.word 1,	0,	2,	1,	2,	0,	1,	1,	3,	3,	2,	2,	1,	2,	1,	2,	1,	1,	2,	2
291	#	.word 1,	2,	1,	1,	1,	1,	2,	2,	3,	2,	0,	1,	3,	0,	1,	3,	1,	0,	3,	0
292	#	.word 2,	2,	1,	2,	2,	3,	2,	0,	3,	3										
293																					
294	#	.word 1,	0,	2,	2,	1,	2,	2,	0,	2,	2					# 30					
295	#	.word 2,	3,	0,	2,	1,	2,	1,	2,	1,	2,	0,	1,	3,	0,	3,	3,	3,	1,	1,	2
296	#	.word 3,	1,	2,	0,	0,	2,	2,	0,	1,	1,	3,	2,	2,	2,	2,	2,	3,	1,	1,	3
297																					
298	#	.word 3,	2,	0,	3,	3,	3,	1,	3,	3,	0,	1,	2,	0,	2,	1,	0,	2,	2,	2,	0
299	#	.word 2,	0,	1,	0,	3,	2,	2,	2,	0,	2,	1,	2,	2,	2,	1,	0,	0,	2,	0,	1
300	#	.word 2,	3,	1,	3,	0,	1,	2,	0,	2,	1										
301																					
302	#	.word 3,	1,	1,	0,	0,	1,	3,	0,	3,	1					# 32					
303	#	.word 1,	3,	3,	0,	0,	2,	1,	1,	3,	0,	2,	3,	1,	2,	0,	2,	3,	1,	2,	3
304	#	.word 2,	0,	0,	1,	3,	2,	3,	1,	1,	0,	2,	1,	2,	3,	1,	2,	3,	3,	2,	2
305																					
306	#	.word 2,	3,	0,	0,	2,	0,	2,	0,	0,	2,	3,	0,	3,	3,	2,	0,	1,	2,	2,	2
307	#	.word 1,	1,	2,	3,	0,	0,	2,	3,	3,	3,	0,	2,	0,	2,	1,	2,	3,	1,	3,	1
308	#	.word 2,	2,	3,	0,	1,	0,	2,	3,	0,	3										
309																					
310	#	.word 0,	0,	1,	2,	2,	3,	1,	1,	3,	3					# 34					
311	#	.word 2,	2,	0,	2,	2,	1,	3,	3,	2,	3,	2,	2,	2,	0,	2,	0,	0,	1,	2,	3
312	#	.word 2,	0,	1,	0,	0,	0,	1,	3,	3,	0,	1,	1,	1,	1,	0,	3,	1,	1,	3,	3
313																					
314	#	.word 1,	3,	0,	3,	0,	0,	0,	3,	1,	0,	2,	2,	3,	1,	2,	0,	2,	0,	2,	3
315	#	.word 2,	2,	1,	3,	3,	0,	3,	0,	3,	3,	3,	2,	0,	0,	0,	2,	0,	2,	3,	0
316	#	.word 2,	2,	1,	1,	3,	0,	0,	0,	0,	0										
317																					
318	#	.word 2,	0,	2,	2,	1,	2,	2,	2,	3,	3					# 36					
319	#	.word 3,	0,	1,	2,	2,	2,	2,	2,	0,	3,	2,	1,	3,	0,	1,	2,	0,	0,	0,	1
320	#	.word 0,	0,	0,	3,	1,	2,	0,	2,	0,	3,	3,	3,	3,	0,	0,	3,	1,	2,	1,	3
321																					
322	#	.word 3,	0,	2,	0,	0,	2,	3,	0,	2,	2,	1,	3,	2,	1,	3,	1,	3,	2,	0,	2
323	#	.word 1,	0,	3,	0,	1,	0,	0,	2,	1,	3,	2,	1,	2,	0,	3,	1,	2,	2,	0,	3
324	#	.word 2,	3,	0,	2,	2,	2,	1,	1,	0,	1										
325																					
326	#	.word 1,	2,	0,	1,	0,	2,	3,	0,	2,	1					# 38					
327	#	.word 3,	2,	1,	1,	3,	2,	1,	2,	2,	0,	1,	3,	3,	3,	0,	1,	1,	2,	2,	3
328	#	.word 3,	1,	1,	3,	1,	2,	3,	1,	0,	2,	3,	3,	3,	0,	3,	0,	3,	3,	2,	0
329																					
330	#	.word 1,	2,	0,	0,	0,	0,	2,	1,	3,	2,	0,	2,	0,	1,	2,	1,	3,	1,	1,	2
331	#	.word 1,	0,	3,	0,	2,	1,	2,	2,	0,	0,	0,	0,	0,	0,	1,	0,	1,	3,	1,	1
332	#	.word 3,	1,	0,	3,	1,	2,	0,	0,	0,	1										
333																					

346 # .word 0, 1, 2, 3, 0, 2, 1, 0, 3, 1, 1, 0, 3, 1, 1, 3, 0, 1, 2, 3
347 # .word 0, 2, 2, 2, 1, 0, 2, 0, 3, 3, 3, 3, 3, 1, 2, 3, 0, 3, 2, 0, 1
348 # .word 2, 2, 3, 0, 1, 3, 1, 0, 1, 0
349
350 # .word 3, 0, 2, 2, 3, 0, 0, 0, 2, 1 # 44
351 # .word 3, 3, 1, 1, 1, 2, 2, 1, 3, 1, 2, 2, 2, 2, 0, 0, 3, 1, 2, 2
352 # .word 2, 0, 3, 2, 3, 0, 2, 0, 0, 0, 1, 3, 2, 0, 1, 0, 2, 1, 2, 2
353
354 # .word 1, 2, 0, 0, 3, 1, 2, 0, 1, 2, 1, 0, 0, 3, 2, 1, 3, 3, 2, 1
355 # .word 1, 3, 1, 0, 1, 1, 1, 3, 2, 3, 2, 2, 2, 1, 0, 2, 1, 3, 1, 0
356 # .word 3, 1, 3, 0, 0, 2, 3, 1, 2, 1
357
358 # .word 0, 0, 3, 3, 3, 3, 1, 1, 3, 0 # 46
359 # .word 2, 0, 3, 2, 3, 0, 1, 2, 2, 3, 1, 0, 0, 1, 3, 3, 2, 0, 3, 2
360 # .word 3, 3, 2, 2, 3, 0, 1, 0, 0, 3, 0, 2, 3, 3, 3, 1, 3, 0, 0, 2
361
362 # .word 1, 1, 2, 1, 0, 1, 2, 1, 3, 3, 0, 0, 2, 0, 3, 2, 1, 2, 1, 3
363 # .word 0, 2, 2, 0, 3, 0, 2, 3, 0, 0, 3, 1, 1, 3, 3, 0, 1, 0, 0, 3
364 # .word 3, 0, 3, 3, 0, 3, 0, 1, 3, 1
365
366 # .word 1, 3, 3, 3, 0, 0, 3, 2, 3, 0 # 48
367 # .word 0, 2, 1, 1, 3, 2, 1, 0, 1, 1, 2, 0, 0, 1, 2, 3, 0, 3, 2, 3
368 # .word 1, 3, 0, 3, 2, 2, 0, 3, 1, 2, 1, 1, 1, 0, 3, 0, 1, 2, 0, 2
369
370 # .word 1, 3, 2, 3, 1, 2, 2, 0, 2, 2, 1, 1, 2, 0, 1, 2, 1, 0, 2, 0
371 # .word 2, 1, 0, 1, 1, 2, 0, 2, 3, 0, 2, 2, 0, 2, 2, 1, 2, 2, 3, 3
372 # .word 2, 2, 2, 1, 1, 0, 1, 3, 1, 1
373
374 # .word 2, 2, 1, 2, 0, 0, 0, 1, 0, 2 # 50
375 # .word 2, 0, 2, 3, 2, 3, 1, 1, 1, 3, 2, 0, 0, 0, 1, 3, 3, 2, 1, 1
376 # .word 3, 2, 3, 1, 2, 1, 2, 1, 2, 1, 3, 2, 3, 2, 3, 3, 2, 1, 3, 0
377
378 #B:
379 # .word 1, 1, 0, 3, 1, 2, 0, 0, 0, 0, 0, 2, 1, 2, 3, 0, 0, 3, 3, 2
380 # .word 2, 1, 2, 3, 3, 0, 2, 2, 1, 1, 2, 2, 0, 2, 2, 1, 2, 3, 2, 2
381 # .word 3, 3, 2, 2, 1, 1, 1, 1, 2, 1
382
383 # .word 2, 2, 3, 3, 3, 0, 0, 3, 2, 3 # 2
384 # .word 2, 3, 1, 2, 1, 1, 2, 2, 0, 1, 0, 3, 2, 1, 1, 1, 2, 0, 1, 2
385 # .word 2, 0, 2, 1, 3, 3, 2, 3, 2, 0, 3, 1, 3, 3, 2, 0, 1, 0, 1, 1
386
387 # .word 2, 2, 1, 1, 2, 2, 1, 2, 3, 3, 1, 3, 2, 2, 2, 3, 3, 1, 0, 2
388 # .word 1, 0, 0, 0, 1, 1, 2, 0, 3, 2, 3, 3, 0, 2, 3, 1, 0, 0, 2, 1
389 # .word 2, 0, 2, 1, 1, 2, 3, 1, 3, 2
390
391 # .word 1, 0, 0, 0, 0, 0, 2, 2, 0, 2 # 4
392 # .word 1, 2, 0, 3, 2, 2, 0, 0, 3, 2, 1, 1, 3, 0, 2, 0, 0, 1, 0, 2
393 # .word 3, 3, 1, 3, 3, 0, 0, 2, 2, 0, 0, 0, 1, 0, 0, 1, 3, 0, 2, 1
394
395 # .word 3, 2, 2, 1, 3, 2, 0, 1, 2, 2, 3, 2, 1, 1, 1, 1, 3, 0, 1, 3
396 # .word 2, 2, 3, 1, 1, 2, 0, 2, 1, 1, 2, 3, 1, 0, 1, 0, 1, 1, 0, 0
397 # .word 2, 0, 3, 0, 3, 0, 3, 2, 2, 3
398
399 # .word 3, 2, 1, 0, 2, 2, 1, 1, 0, 3 # 6
400 # .word 3, 2, 2, 0, 0, 3, 0, 1, 0, 0, 1, 2, 0, 1, 3, 0, 1, 2, 2, 0
401 # .word 0, 3, 0, 3, 0, 1, 1, 2, 0, 0, 0, 3, 0, 0, 2, 1, 1, 1, 0, 2
402
403 # .word 1, 3, 1, 2, 0, 3, 0, 3, 1, 3, 0, 0, 2, 2, 2, 2, 3, 3, 2, 1
404 # .word 2, 2, 1, 1, 2, 2, 2, 2, 0, 3, 0, 0, 2, 0, 1, 2, 0, 3, 2, 3
405 # .word 2, 0, 2, 1, 2, 1, 0, 2, 1, 1
406
407 # .word 3, 2, 2, 3, 1, 0, 3, 3, 1, 0 # 8
408 # .word 3, 2, 2, 0, 0, 3, 0, 0, 2, 0, 3, 2, 3, 1, 1, 0, 0, 2, 3, 0
409 # .word 0, 1, 1, 1, 2, 1, 3, 2, 1, 3, 0, 1, 3, 3, 1, 1, 1, 1, 1, 1
410
411 # .word 0, 0, 2, 3, 2, 2, 2, 3, 2, 3, 1, 2, 3, 2, 2, 2, 0, 1, 3, 0
412 # .word 1, 1, 0, 1, 0, 1, 1, 3, 3, 1, 2, 2, 3, 2, 0, 2, 2, 0, 1, 3
413 # .word 0, 1, 3, 2, 1, 3, 3, 2, 0, 1
414
415 # .word 3, 2, 0, 2, 1, 1, 0, 3, 0, 1 # 10
416 # .word 1, 1, 1, 1, 3, 0, 0, 1, 0, 2, 3, 1, 3, 0, 2, 1, 3, 0, 3, 0
417 # .word 3, 2, 2, 0, 0, 2, 1, 3, 3, 2, 3, 2, 2, 1, 2, 2, 3, 0, 3, 2
418
419 # .word 2, 0, 3, 2, 3, 2, 0, 0, 1, 2, 0, 0, 2, 0, 0, 3, 3, 2, 0, 0
420 # .word 3, 3, 0, 2, 3, 1, 0, 1, 0, 2, 1, 0, 2, 1, 0, 1, 0, 3, 0, 2
421 # .word 2, 3, 0, 0, 2, 1, 0, 1, 0, 0
422
423 # .word 0, 2, 2, 3, 2, 0, 3, 3, 2, 1 # 12
424 # .word 0, 0, 3, 1, 2, 3, 3, 1, 0, 3, 1, 1, 0, 3, 3, 3, 2, 2, 2, 0
425 # .word 1, 2, 0, 3, 0, 1, 0, 1, 1, 0, 1, 2, 0, 3, 2, 0, 1, 2, 2, 0
426
427 # .word 2, 0, 0, 1, 0, 3, 0, 3, 2, 1, 1, 1, 1, 3, 2, 1, 1, 1, 1, 0
428 # .word 2, 1, 1, 3, 2, 0, 2, 1, 1, 0, 2, 2, 1, 3, 0, 2, 1, 0, 1, 2
429 # .word 0, 1, 3, 2, 3, 2, 1, 0, 2, 0
430
431 # .word 2, 2, 3, 1, 1, 3, 2, 3, 2, 2 # 14
432 # .word 0, 2, 0, 0, 0, 3, 2, 0, 2, 2, 3, 3, 3, 2, 1, 2, 0, 0, 3, 0
433 # .word 2, 0, 3, 2, 2, 3, 0, 3, 2, 1, 2, 2, 1, 2, 0, 0, 3, 1, 2, 0
434
435 # .word 2, 3, 2, 2, 1, 1, 1, 3, 3, 3, 3, 3, 1, 3, 0, 1, 3, 2, 2, 1
436 # .word 0, 1, 1, 2, 1, 2, 3, 1, 2, 2, 1, 2, 1, 1, 0, 3, 3, 1, 1, 3

437	#	.word	2,	0,	0,	1,	2,	0,	1,	3,	1,	0																		
438																														
439	#	.word	0,	2,	2,	3,	3,	0,	2,	3,	2,	1																		
440	#	.word	1,	3,	0,	2,	2,	3,	3,	1,	2,	3,	3,	3,	1,	3,	0,	3,	1,	1,	2,	2								
441	#	.word	2,	1,	0,	3,	2,	3,	0,	2,	3,	2,	3,	1,	2,	3,	3,	1,	2,	1,	0,	0								
442																														
443	#	.word	0,	3,	3,	3,	3,	0,	3,	3,	3,	3,	2,	1,	0,	3,	0,	3,	2,	3,	1,	0	3							
444	#	.word	0,	1,	3,	1,	0,	2,	2,	3,	1,	0,	2,	1,	1,	3,	1,	1,	3,	1,	2,	1								
445	#	.word	0,	0,	3,	2,	1,	1,	1,	1,	3,	2																		
446																														
447	#	.word	1,	3,	3,	1,	0,	3,	1,	1,	2,	0																		
448	#	.word	0,	0,	2,	3,	3,	2,	2,	3,	0,	2,	3,	1,	3,	3,	0,	2,	1,	2,	2,	2								
449	#	.word	1,	0,	1,	3,	2,	3,	1,	1,	2,	1,	1,	0,	0,	2,	3,	2,	1,	0,	3,	1								
450																														
451	#	.word	3,	0,	1,	1,	2,	2,	1,	3,	3,	1,	1,	0,	0,	3,	3,	0,	0,	0,	0,	0	3							
452	#	.word	3,	1,	3,	0,	0,	0,	3,	3,	2,	1,	3,	0,	1,	3,	1,	1,	1,	0,	1,	0								
453	#	.word	1,	2,	2,	2,	3,	3,	0,	2,	3,	2																		
454																														
455	#	.word	1,	3,	3,	1,	1,	3,	0,	3,	3,	2																		
456	#	.word	1,	1,	2,	0,	3,	0,	1,	2,	1,	1,	0,	0,	1,	2,	2,	0,	3,	1,	1,	1								
457	#	.word	3,	3,	3,	1,	0,	3,	3,	2,	2,	2,	1,	2,	0,	1,	1,	3,	0,	3,	1,	0								
458																														
459	#	.word	2,	2,	0,	1,	2,	3,	2,	1,	2,	0,	3,	2,	1,	3,	0,	1,	2,	0,	3,	0	3							
460	#	.word	1,	1,	2,	1,	3,	2,	3,	0,	2,	1,	1,	3,	0,	2,	3,	2,	1,	3,	2,	2								
461	#	.word	3,	2,	1,	2,	3,	0,	3,	0,	2,	1																		
462																														
463	#	.word	3,	1,	2,	1,	0,	2,	0,	1,	2,	2																		
464	#	.word	0,	1,	1,	2,	0,	2,	0,	0,	2,	1,	3,	1,	1,	0,	3,	1,	2,	1,	3,	2								
465	#	.word	2,	3,	3,	1,	0,	1,	3,	1,	1,	3,	1,	2,	1,	0,	1,	2,	0,	3,	1,	3								
466																														
467	#	.word	3,	2,	0,	2,	0,	2,	3,	0,	0,	1,	2,	2,	0,	0,	0,	1,	2,	2,	1,	3	3							
468	#	.word	0,	2,	1,	0,	3,	0,	3,	1,	0,	0,	0,	3,	0,	3,	2,	1,	0,	3,	3,	3								
469	#	.word	2,	1,	3,	3,	1,	2,	1,	1,	0,	2																		
470																														
471	#	.word	0,	0,	2,	2,	2,	0,	0,	0,	3,	3																		
472	#	.word	0,	1,	2,	1,	1,	1,	1,	1,	2,	1,	1,	2,	3,	0,	2,	2,	0,	1,	0,	0								
473	#	.word	1,	0,	1,	1,	3,	3,	2,	0,	0,	3,	3,	1,	3,	0,	2,	0,	1,	1,	0,	0								
474																														
475	#	.word	0,	3,	0,	2,	1,	2,	1,	0,	0,	2,	3,	1,	1,	0,	2,	3,	2,	0,	1,	1	3							
476	#	.word	0,	3,	0,	2,	0,	1,	0,	1,	0,	2,	2,	0,	3,	3,	2,	1,	0,	3,	0,	1								
477	#	.word	1,	0,	1,	0,	3,	1,	2,	0,	0,	1																		
478																														
479	#	.word	1,	0,	1,	0,	1,	2,	2,	0,	3,	0																		
480	#	.word	0,	3,	0,	3,	2,	2,	1,	0,	0,	3,	3,	3,	0,	1,	1,	0,	0,	1,	2,	0								
481	#	.word	2,	1,	3,	0,	3,	3,	3,	2,	1,	0,	1,	2,	3,	1,	1,	3,	2,	3,	0,	2								
482																														
483	#	.word	3,	3,	0,	1,	0,	3,	2,	3,	0,	2,	1,	3,	1,	1,	1,	1,	0,	3,	3,	0	3							
484	#	.word	1,	3,	3,	2,	2,	3,	2,	0,	2,	1,	3,	0,	0,	3,	2,	3,	2,	1,	2,	3								
485	#	.word	1,	1,	1,	3,	2,	0,	1,	0,	0,	3																		
486																														
487	#	.word	3,	3,	3,	0,	2,	3,	1,	0,	3,	1																		
488	#	.word	3,	1,	1,	0,	0,	2,	1,	2,	2,	1,	2,	0,	3,	0,	3,	0,	3,	3,	2,	3								
489	#	.word	2,	3,	1,	1,	2,	3,	3,	3,	0,	1,	3,	0,	1,	2,	0,	1,	3,	2,	2,	2								
490																														
491	#	.word	2,	1,	3,	3,	2,	3,	3,	0,	3,	3,	3,	3,	0,	1,	2,	0,	3,	0,	0,	1	3							
492	#	.word	0,	3,	0,	1,	1,	2,	0,	1,	0,	2,	0,	1,	0,	0,	0,	3,	3,	3,	3,	2								
493	#	.word	0,	2,	1,	1,	0,	1,	0,	3,	2,	0																		
494																														
495	#	.word	0,	1,	0,	3,	0,	0,	1,	0,	1,	3																		
496	#	.word	2,	2,	1,	3,	0,	1,	0,	1,	2,	2,	3,	0,	2,	1,	2,	1,	0,	1,	0,	0								
497	#	.word	2,	1,	1,	1,	2,	0,	1,	0,	1,	3,	2,	3,	0,	0,	2,	3,	1,	0,	2,	0								
498																														
499	#	.word	3,	2,	3,	3,	0,	3,	0,	2,	0,	0,	1,	2,	3,	3,	2,	2,	0,	0,	0,	0	3							
500	#	.word	1,	1,	3,	3,	2,	1,	3,	1,	0,	3,	1,	2,	2,	3,	0,	3,	1,	1,	2,	2								
501	#	.word	0,	3,	2,	2,	0,	2,	3,	3,	2,	3																		
502																														
503	#	.word	0,	0,	3,	2,	2,	0,	1,	2,	3,	3																		
504	#	.word	2,	3,	1,	1,	1,	0,	2,	3,	2,	3,	1,	0,	2,	2,	0,	2,	3,	1,	3,	1								
505	#	.word	3,	0,	0,	2,	0,	3,	1,	3,	1,	2,	0,	3,	3,	1,	0,	1,	1,	0,	0,	3								
506																														
507	#	.word	1,	1,	3,	2,	0,	3,	2,	2,	0,	1,	1,	0,	2,	0,	3,	2,	0,	0,	2,	2	3							
508	#	.word	0,	1,	1,	3,	1,	1,	3,	3,	1,	3,	1,	0,	2,	2,	2,	1,	2,	0,	2,	1								
509	#	.word	0,	3,	3,	1,	0,	0,	1,	3,	0,	0																		
510																														
511	#	.word	1,	2,	1,	1,	0,	2,	0,	0,	1,	0																		
512	#	.word	0,	2,	2,	2,	3,	2,	1,	3,	1,	0,	3,	0,	3,	1,	1,	2,	0,	1,	2,	0								
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514																														
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691 # 133, 116, 90, 136, 95, 124, 99, 110, 116, 123, 103, 115, 113, 104, 102, 93, 101, 114, 97, 99,
692 # 130, 130, 130, 117, 106, 141, 107, 127, 118, 119, 104, 123, 82, 125, 88, 118, 107, 94, 105, 95,
693 # 109, 118, 138, 109, 96, 104, 117, 112, 126, 101, 134, 96, 116, 117, 95, 141, 121, 131, 113, 126,
694 # 98, 117, 113, 100, 110, 96, 95, 120, 110, 108, 138, 124, 120, 109, 113, 168, 108, 151, 152, 137,
695 # 136, 141, 108, 129, 120, 145, 106, 106, 105, 91, 107, 127, 133, 121, 88, 113, 125, 111, 116, 120,
696 # 154, 125, 126, 143, 110, 155, 128, 119, 127, 128, 86, 129, 129, 94, 111, 115, 129, 128, 100, 118,
697 # 135, 133, 111, 95, 89, 139, 98, 135, 111, 105, 111, 116, 105, 106, 114, 119, 82, 109, 109, 79,
698 # 104, 99, 115, 93, 89, 94, 142, 105, 103, 114, 143, 115, 107, 143, 102, 146, 118, 122, 121, 127,
699 # 80, 100, 110, 106, 104, 108, 107, 108, 91, 113, 102, 112, 106, 114, 105, 114, 106, 134, 133, 128,
700 # 106, 138, 98, 128, 103, 126, 108, 95, 100, 90, 102, 120, 121, 102, 90, 99, 116, 105, 119, 97,
701 # 149, 112, 108, 124, 114, 141, 104, 126, 118, 122, 105, 100, 125, 110, 91, 106, 123, 104, 95, 92,
702 # 124, 101, 105, 105, 115, 126, 102, 119, 126, 113, 110, 123, 91, 119, 106, 124, 93, 91, 90, 92,
703 # 95, 129, 116, 96, 86, 97, 123, 109, 112, 103, 133, 118, 101, 117, 101, 141, 105, 127, 120, 112,
704 # 90, 111, 120, 108, 94, 98, 114, 102, 105, 102, 135, 128, 123, 116, 116, 148, 115, 149, 134, 112,
705 # 113, 146, 94, 126, 119, 135, 100, 110, 121, 103, 116, 139, 144, 123, 95, 111, 141, 107, 121, 123,
706 # 156, 111, 127, 143, 123, 165, 127, 139, 138, 132, 101, 138, 121, 114, 111, 110, 117, 124, 99, 98
707
708 # ~~~~~ text segment ~~~~~
709 .text
```

```

710 .global main          # main func is accessed by all
711
712 main:
713     la s0, A           # s0 is the base addr of A
714     la s1, B           # s1 is the base addr of B
715     la s2, C           # s2 is the base addr of C
716
717     # ~~~~ caller (main) saves non-preserved ~~~~
718     # ~~ 1. Make space on stack for 8 registers (main:caller) ~~
719     addi sp, sp, -32
720
721     # ~~ 2. Store reg values in memory (main:caller) ~~
722     sw a0, 28(sp)
723     sw a1, 24(sp)
724     sw a2, 20(sp)
725     sw a3, 16(sp)
726     sw a4, 12(sp)
727     sw a5, 8(sp)
728     sw a6, 4(sp)
729     sw a7, 0(sp)
730
731     # ~~ 3. Execute function (main:caller) ~~
732     addi a0, s0, 0      # a0 is base addr of mat A
733     addi a1, s1, 0      # a1 is base addr of mat B
734     addi a2, s2, 0      # a2 is base addr of mat C
735     addi a3, zero, M    # a3 is one dim of the mat
736     jal matmul          # no return val: C is in mem already
737
738     # ~~ 4. Restore og values from stack (main:caller) ~~
739     lw a0, 28(sp)
740     lw a1, 24(sp)
741     lw a2, 20(sp)
742     lw a3, 16(sp)
743     lw a4, 12(sp)
744     lw a5, 8(sp)
745     lw a6, 4(sp)
746     lw a7, 0(sp)
747
748     # ~~ 5. Deallocate space on stack (main:caller) ~~
749     addi sp, sp, 32
750
751     # ~~ end program (main:caller) ~~
752     j end
753
754 matmul:
755     # ~~~~ callee (matmul) saves preserved ~~~~
756     # ~~ 1. Make space on stack for 13 registers (matmul:callee) ~~
757     addi sp, sp, -52
758
759     # ~~ 2. Store reg values in memory (matmul:callee) ~~
760     sw s0, 48(sp)
761     sw s1, 44(sp)
762     sw s2, 40(sp)
763     sw s3, 36(sp)
764     sw s4, 32(sp)
765     sw s5, 28(sp)
766     sw s6, 24(sp)
767     sw s7, 20(sp)
768     sw s8, 16(sp)
769     sw s9, 12(sp)
770     sw s10, 8(sp)
771     sw s11, 4(sp)
772     sw ra, 0(sp)        # preserve ra to return to main
773
774     # ~~ 3. Execute function (matmul:callee) ~~
775
776     # ~ init regs ~
777         # (given) a0 is base addr for A
778         # (given) a1 is base addr for B
779         # (given) a2 is base addr for C
780         # (given) a3 is one dim of mat (loop limit)
781
782     addi s0, zero, 0     # s0 is 'i', the row idx of A
783     addi s1, zero, 0     # s1 is 'j', the col idx of A
784     addi s2, zero, 0     # s2 is 'k', the working idx of A and B
785
786         # s3 is reserved for A[i][k]
787         # s4 is reserved for B[k][j]
788         # s5 is reserved for C[i][j]
789     sub s6, zero, a3     # s6 is reverved for (i*m) = -m offset
790     sub s7, zero, a3     # s7 is reserved for (k*m) = -m offset
791
792         # s8 = sum = dot prod of C[i][j]
793
794     # ~ for all rows in A ~
795 for_row: bge s0, a3, endfor_row # i >= m ?
796
797     add s6, s6, a3        #s6 = (i*m)
798         # neg offset makes 0 at start,
799         #   than add m each iter
800

```

```

801 # ~ for all columns in A ~
802 for_col: bge s1, a3, endfor_col # j >= m ?
803
804 # load addr (C + 4*(i*m + j))
805 addi s5, s6, 0 # s5 = i*m
806 add s5, s5, s1 # s5 = i*m + j
807 slli s5, s5, 2 # s5 = 4*(i*m + j)
808 add s5, s5, a2 # s5 = C + 4*(i*m + j)
809
810 addi s8, zero, 0 # s8 = sum = 0
811
812
813 # ~ get the dot prod of A,B at (i, j) ~
814 for_dot: bge s2, a3, endfor_dot # k >= m ?
815
816 add s7, s7, a3 # s7 = (k*m)
817 # neg offset makes 0 at start,
818 # than add m each iter
819
820 # load A[i][k] = *(A + 4*(i*m + k))
821 addi s3, s6, 0 # s3 = i*m
822 add s3, s3, s2 # s3 = i*m + k
823 slli s3, s3, 2 # s3 = 4*(i*m + k)
824 add s3, s3, a0 # s3 = A + 4*(i*m + k)
825 lw s3, 0(s3) # s3 = *(A + 4*(i*m + k)) = A[i][k]
826
827 # load B[k][j]= *(B + 4*(k*m + j))
828 addi s4, s7, 0 # s4 = k*m
829 add s4, s4, s1 # s4 = k*m + j
830 slli s4, s4, 2 # s4 = 4*(k*m + j)
831 add s4, s4, a1 # s4 = B + 4*(k*m + j)
832 lw s4, 0(s4) # s4 = *(B + 4*(k*m + j)) = B[k][j]
833
834 # sum += (Aik * Bkj)
835
836 # ---- caller (matmul) saves non-preserved ----
837 # -- 1. Make space on stack for 4 registers (matmul:caller)--
838 addi sp, sp, -20
839
840 # -- 2. Store reg values in memory (matmul:caller)--
841 sw a0, 16(sp)
842 sw a1, 12(sp)
843 sw a2, 8(sp)
844 sw a3, 4(sp)
845 sw ra, 0(sp)
846
847 # -- 3. Execute function (matmul:caller)--
848 addi a0, s3, 0 # a0 is A[i][k]
849 addi a1, s4, 0 # a1 is B[k][j] # here a1 goes neg
850 jal multiply
851 add s8, s8, a0 # sum += A[i][k] * B[k][j]
852
853 # -- 4. Restore og values from stack (matmul:caller)--
854 lw a0, 16(sp)
855 lw a1, 12(sp)
856 lw a2, 8(sp)
857 lw a3, 4(sp)
858 lw ra, 0(sp)
859
860 # -- 5. Deallocate space on stack (matmul:caller)--
861 addi sp, sp, 20
862
863 # increment and return
864 addi s2, s2, 1 # k++
865 j for_dot # loop back
866
867 endfor_dot:
868 sw s8, 0(s5) # C[i][j] = sum
869 addi s2, zero, 0 # reset k = 0
870 sub s7, zero, a3 # reset k*m
871 addi s1, s1, 1 # j++
872 j for_col
873
874 endfor_col:
875 addi s1, zero, 0 # reset j = 0
876 addi s0, s0, 1 # i++
877 j for_row
878
879 endfor_row:
880
881 # ~~ 4. Restore og values from stack (matmul:callee) ~~
882 lw s0, 48(sp)
883 lw s1, 44(sp)
884 lw s2, 40(sp)
885 lw s3, 36(sp)
886 lw s4, 32(sp)
887 lw s5, 28(sp)
888 lw s6, 24(sp)
889 lw s7, 20(sp)
890 lw s8, 16(sp)
891 lw s9, 12(sp)

```

```

892 lw s10, 8(sp)
893 lw s11, 4(sp)
894 lw ra, 0(sp)
895
896 # ~~ 5. Deallocate space on stack (matmul:callee)~~
897 addi sp, sp, 52
898
899 # ~~ exit func (matmul:callee) ~~
900 jr ra
901
902
903
904 multiply:
905
906 # ~~~~ callee (multiply) saves preserved ~~~~
907 # but since we dont write to preseved, we dont need to save regs
908
909 # ~~ init vars ~~
910 addi t0, zero, 0 # b goes to 0
911 addi t1, a0, 0 # t1 = a (const)
912 addi a0, zero, 0 # let a0 = rolling product (easy return)
913 # let a1 = b
914
915 while: bge t0, a1 endwhile #while b > 0
916 add a0, a0, t1 # prod += a
917 addi a1, a1, -1 # b--
918 j while
919
920 endwhile:
921 # ~~ return with return register (+4) ~~
922 jr ra
923
924
925
926
927
928 end: # OTTER specific ending
929 #addi a0, zero, 10
930 # ecall

```

0.2 Diff Shell Script

Listing 2: test.sh

```

1 #!/bin/bash
2
3
4 diff ./datasets/d3_soln.txt ./asm_outputs/output_TC3.txt
5 diff ./datasets/d10_soln.txt ./asm_outputs/output_TC10.txt
6 diff ./datasets/d16_soln.txt ./asm_outputs/output_TC16.txt
7 diff ./datasets/d50_soln.txt ./asm_outputs/output_TC50.txt

```

Test Cases

0.3 Validation Process

Validating the matrix multiplication was a three step process.

1. Writing the shell script to DIFF the output and solution files. See [0.2](#).
2. Processing the solution matrix to match RARS output, building the solution into a list of 8 byte hex values with uppercase alphabetic letters. This was done in vim using the commands :
 - Convert the decimal values to hex and prefix with leading zeros to reach 8 bytes.

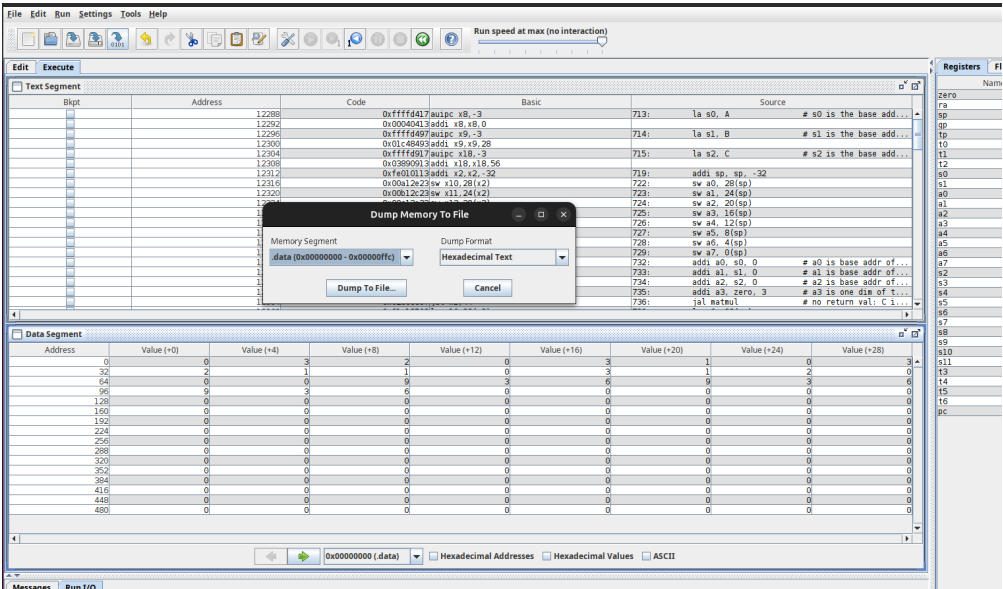
```
:%s/.*\/\=printf("%08X", str2nr(submatch(0)))/g
```
 - Capitalize the alphabet characters.

```
:%s/[A-Za-z]/\=toupper(submatch(0))/g
```

Listing 3: Snippet of datasets/d10_soln.txt

```
1 0000001C
2 0000001F
3 00000023
4 00000023
5 00000024
6 0000000C
7 00000014
8 00000019
9 0000001D
10 00000017
11 0000001E
12 0000001C
13 00000014
14 00000026
15 0000001D
:
:
```

3. Steps to collect RARS data :
 - Memory Config was set to **Compact, Data At Address 0**.
 - Run assembly file and the program
 - Save the output matrix with the **Dump Machine Code** option, where I selected **.data** for the Memory Segment in the **Hexidecimal Text** Dump Format.
 - Go into the text file and manually delete the hex values that are not part of the resulting matrix.



0.4 Results

All DIFFs passed except for the 50×50 matrix. The RISC-V OTTER memory mapping did not have enough memory to store both matrices A, B , throwing a compile time error. Below are the screenshots of the matrices in memory. It should be noted that const M is set to a single dimension of the matrices.

0.4.1 dataset3.h

```
19 | # ~~~~~ compile time ~~~~~
20 |
21 | .equ M, 3 # one dim of the mat
22 |
23 |
24 | # ~~~~~ data segment (memory) ~~~~~
25 | .data
26 |
27 |
28 |
29 |
30 |
31 |
32 |
33 |
34 |
35 |
36 |
37 |
38 |
39 |
40 |
41 |
42 |
43 |
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72 |
73 |
74 |
75 |
76 |
77 |
78 |
79 |
80 |
81 |
82 |
83 | # ~~ TC : dataset3.h ~~
84 | A:      .word 0, 3, 2, 0, 3, 1, 0, 3, 2
85 | B:      .word 1, 1, 0, 3, 1, 2, 0, 0, 0
86 | C:      .word -7:9
87 | #answer = 9, 3, 6, 9, 3, 6, 9, 3, 6
88 |
89 |
```

Data Segment									
Address	Value (+0)	Value (+4)	Value (+8)	Value (+12)	Value (+16)	Value (+20)	Value (+24)	Value (+28)	
0	0	3	2	0	3	1	0	3	
32	2	1	1	0	3	1	2	0	
64	0	0	9	3	6	9	3	6	
96	9	3	6	0	0	0	0	0	
128	0	0	0	0	0	0	0	0	
160	0	0	0	0	0	0	0	0	
192	0	0	0	0	0	0	0	0	
224	0	0	0	0	0	0	0	0	
256	0	0	0	0	0	0	0	0	
288	0	0	0	0	0	0	0	0	
320	0	0	0	0	0	0	0	0	
352	0	0	0	0	0	0	0	0	
384	0	0	0	0	0	0	0	0	
416	0	0	0	0	0	0	0	0	
448	0	0	0	0	0	0	0	0	
480	0	0	0	0	0	0	0	0	

0.4.2 dataset10.h

```
19 | # ~~~~~ compile time ~~~~~
20 |
21 | .equ M, 10 # one dim of the mat
22 |
23 |
24 | # ~~~~~ data segment (memory) ~~~~~
25 |
26 |
27 |
28 |
29 |
30 |
31 |
32 |
33 |
34 |
35 |
36 |
37 |
38 |
39 |
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74 |
75 |
76 |
77 |
78 |
79 |
80 |
81 |
82 |
83 |
84 |
85 |
86 |
87 |
88 |
89 | # ~~ TC : dataset10.h ~~
90 |
91 | A:      .word 0, 3, 2, 0, 3, 1, 0, 3, 2, 3 #1
92 |          .word 2, 0, 3, 3, 1, 2, 3, 0, 0, 1 #2
93 |          .word 1, 1, 2, 3, 1, 2, 3, 1, 1, 3 #3
94 |          .word 2, 2, 0, 1, 3, 2, 2, 2, 0, 0 #4
95 |          .word 1, 0, 1, 3, 3, 0, 3, 3, 3, 3 #5
96 |          .word 0, 3, 2, 1, 2, 2, 0, 0, 3, 0 #6
97 |          .word 1, 1, 0, 3, 3, 1, 2, 3, 3, 0 #7
98 |          .word 1, 2, 1, 0, 1, 2, 2, 1, 0, 3 #8
99 |          .word 1, 0, 2, 2, 1, 1, 1, 1, 1, 1 #9
100 |         .word 2, 0, 3, 1, 1, 2, 2, 3, 3, 1 #10
101 |
102 |
103 | B:      .word 1, 1, 0, 3, 1, 2, 0, 0, 0, 0 #1
104 |          .word 0, 2, 1, 2, 3, 0, 0, 3, 3, 2 #2
105 |          .word 2, 1, 2, 3, 3, 0, 2, 2, 1, 1 #3
106 |          .word 2, 2, 0, 2, 2, 1, 2, 3, 2, 2 #4
107 |          .word 3, 3, 2, 2, 1, 1, 1, 1, 2, 1 #5
108 |          .word 2, 2, 3, 3, 3, 0, 0, 3, 2, 3 #6
109 |          .word 2, 3, 1, 2, 1, 1, 2, 2, 0, 1 #7
110 |          .word 0, 3, 2, 1, 1, 1, 2, 0, 1, 2 #8
111 |          .word 2, 0, 2, 1, 3, 3, 2, 3, 2, 0 #9
112 |          .word 3, 1, 3, 3, 2, 0, 1, 0, 1, 1 #10
113 |
114 |
115 |
116 | C:      .word -7:100
117 |
```

```

119
120 #answer =      28, 31, 35, 35, 36, 12, 20, 25, 29, 23 #1
121 #             30, 28, 20, 38, 29, 11, 20, 28, 16, 20 #2
122 #             35, 33, 29, 42, 36, 13, 24, 32, 23, 25 #3
123 #             21, 33, 20, 30, 23, 12, 13, 22, 20, 21 #4
124 #             39, 38, 32, 39, 34, 23, 32, 29, 25, 22 #5
125 #             22, 20, 23, 27, 34, 12, 14, 33, 27, 18 #6
126 #             28, 35, 24, 30, 30, 22, 25, 31, 26, 22 #7
127 #             23, 25, 25, 32, 26, 6, 12, 19, 17, 19  #8
128 #             21, 19, 17, 25, 22, 10, 16, 19, 14, 14 #9
129 #             30, 30, 31, 38, 36, 20, 26, 29, 21, 21 #10
130
131

```

Data Segment									
Address	Value (+0)	Value (+4)	Value (+8)	Value (+12)	Value (+16)	Value (+20)	Value (+24)	Value (+28)	
768	3	3	2	0	1	0	1	1	
800	28	31	35	35	36	12	20	25	
832	25	23	30	28	20	38	29	11	
864	20	28	16	20	35	33	29	42	
896	36	13	24	32	23	25	21	33	
928	20	30	23	12	13	22	20	21	
960	39	38	32	39	34	23	32	29	
992	25	22	22	20	23	27	34	12	
1024	14	33	27	18	28	35	24	30	
1056	30	22	25	31	26	22	23	25	
1088	25	32	26	6	12	19	17	19	
1120	21	19	17	25	22	10	16	19	
1152	14	14	30	30	31	38	36	20	
1184	26	29	21	21	0	0	0	0	
1216	0	0	0	0	0	0	0	0	
1248	0	0	0	0	0	0	0	0	

0.4.3 dataset16.h

```

18
19 # ~~~~~ compile time ~~~~~
20 .equ M, 16 # one dim of the mat
21
22
23
24 # ~~~~~ data segment (memory) ~~~~~

```

Edit	Execute
matrix_multiply.asm*	
131	
132	# ~~ TC : dataset16.h ~~
133	
134	A:
135	
136	.word 0, 3, 2, 0, 3, 1, 0, 3, 2, 3, 2, 0, 3, 3, 1, 2 #1
137	.word 3, 0, 0, 1, 1, 1, 2, 3, 1, 2, 3, 1, 1, 3, 2, 2 #2
138	.word 0, 1, 3, 2, 2, 2, 0, 0, 1, 0, 1, 3, 3, 0, 3, 3 #3
139	.word 3, 3, 0, 3, 2, 1, 2, 2, 0, 0, 3, 0, 1, 1, 0, 3 #4
140	.word 3, 1, 2, 3, 3, 0, 1, 2, 1, 0, 1, 2, 2, 1, 0, 3 #5
141	.word 1, 0, 2, 2, 1, 1, 1, 1, 1, 1, 2, 0, 3, 1, 1, 2 #6
142	.word 2, 3, 3, 1, 3, 2, 0, 0, 0, 3, 3, 3, 2, 1, 2, 3 #7
143	.word 1, 0, 0, 0, 0, 1, 2, 2, 1, 1, 3, 3, 3, 1, 1, 2 #8
144	.word 3, 1, 3, 3, 2, 3, 2, 1, 2, 3, 0, 2, 2, 1, 1, 0 #9
145	.word 0, 0, 0, 0, 1, 3, 3, 1, 1, 1, 2, 2, 3, 2, 1, 1 #10
146	.word 1, 1, 3, 0, 2, 2, 1, 3, 2, 1, 2, 2, 1, 3, 1, 3 #11
147	.word 1, 3, 2, 3, 1, 2, 1, 3, 2, 2, 0, 1, 0, 0, 1, 2 #12
148	.word 3, 3, 1, 0, 0, 0, 3, 1, 2, 3, 2, 3, 2, 0, 0, 0 #13
149	.word 0, 0, 3, 1, 3, 0, 0, 0, 3, 1, 1, 1, 1, 2, 1, 2 #14
150	.word 3, 2, 0, 0, 2, 2, 3, 0, 3, 0, 0, 3, 0, 3, 1, 3 #15
151	.word 3, 1, 1, 1, 2, 2, 1, 3, 0, 3, 3, 1, 0, 0, 3, 2 #16
152	
153	
154	

Edit	Execute
matrix_multiply.asm*	
155	
156	B:
157	
158	.word 1, 1, 0, 3, 1, 2, 0, 0, 0, 0, 0, 2, 1, 2, 3, 0 #1
159	.word 0, 3, 3, 2, 2, 1, 2, 3, 3, 0, 2, 2, 1, 1, 2, 2 #2
160	.word 0, 2, 2, 1, 2, 3, 2, 2, 3, 3, 2, 2, 1, 1, 1, 1 #3
161	.word 2, 1, 2, 2, 3, 3, 3, 0, 0, 3, 2, 3, 2, 3, 1, 2 #4
162	.word 1, 1, 2, 2, 0, 1, 0, 3, 2, 1, 1, 1, 2, 0, 1, 2 #5
163	.word 2, 0, 2, 1, 3, 3, 2, 3, 2, 0, 3, 1, 3, 3, 2, 0 #6
164	.word 1, 0, 1, 1, 2, 2, 1, 1, 2, 2, 1, 2, 3, 3, 1, 3 #7
165	.word 2, 2, 2, 3, 3, 1, 0, 2, 1, 0, 0, 0, 1, 1, 2, 0 #8
166	.word 3, 2, 3, 3, 0, 2, 3, 1, 0, 0, 2, 1, 2, 0, 2, 1 #9
167	.word 1, 2, 3, 1, 3, 2, 1, 0, 0, 0, 0, 2, 2, 2, 0, 2 #10
168	.word 1, 2, 0, 3, 2, 2, 0, 0, 3, 2, 1, 1, 3, 0, 2, 0 #11
169	.word 0, 1, 0, 2, 3, 3, 1, 3, 3, 0, 0, 2, 2, 0, 0, 0 #12
170	.word 1, 0, 0, 1, 3, 0, 2, 1, 3, 2, 2, 1, 3, 2, 0, 1 #13
171	.word 2, 2, 3, 2, 1, 1, 1, 1, 3, 0, 1, 3, 2, 2, 3, 1 #14
172	.word 1, 2, 0, 2, 1, 1, 2, 3, 1, 0, 1, 0, 1, 1, 0, 0 #15
173	.word 2, 0, 3, 0, 3, 0, 3, 2, 2, 3, 3, 2, 1, 0, 2, 2 #16
174	
175	
176	
177	C: .word -7:256 # 16^2 = 256
178	

EditExecute

matrix_multiply.asm*

179

180

181

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199

200

#answer =

#

#

#

#

#

#

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#

#

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#

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#

#

#

#

#

36, 44, 57, 50, 54, 36, 38, 46, 55, 25, 38, 34, 51, 30, 40, 32

37, 34, 38, 52, 51, 40, 28, 32, 41, 22, 26, 35, 49, 35, 42, 23

26, 26, 33, 36, 52, 40, 45, 49, 50, 34, 41, 35, 44, 25, 23, 23

31, 29, 39, 46, 50, 36, 31, 32, 42, 32, 34, 41, 44, 33, 43, 30

31, 28, 39, 46, 50, 40, 35, 37, 43, 35, 33, 43, 43, 29, 37, 29

27, 22, 30, 33, 43, 31, 32, 25, 36, 31, 31, 29, 40, 28, 26, 22

29, 42, 48, 51, 65, 52, 43, 54, 63, 34, 42, 44, 56, 33, 38, 32

26, 22, 23, 38, 49, 32, 26, 30, 43, 22, 24, 27, 45, 24, 26, 17

35, 35, 47, 51, 59, 59, 43, 42, 43, 28, 37, 43, 56, 48, 36, 32

28, 19, 28, 34, 46, 34, 28, 34, 45, 20, 29, 28, 50, 32, 26, 21

37, 38, 51, 50, 55, 45, 38, 49, 56, 28, 38, 40, 50, 29, 44, 26

32, 35, 50, 43, 53, 44, 41, 41, 34, 24, 35, 34, 39, 33, 34, 29

21, 33, 31, 45, 48, 42, 27, 29, 40, 17, 21, 32, 45, 30, 29, 26

26, 27, 38, 33, 29, 31, 32, 31, 35, 25, 29, 29, 34, 15, 25, 23

34, 28, 44, 45, 41, 41, 37, 45, 45, 17, 34, 44, 46, 30, 43, 29

31, 36, 37, 50, 54, 44, 28, 40, 38, 22, 27, 28, 45, 32, 36, 22

Address	Value (+0)	Value (+4)	Value (+8)	Value (+12)	Value (+16)	Value (+20)	Value (+24)	Value (+28)
2048	36	44	57	50	54	36	38	46
2080	55	25	38	34	51	30	40	32
2112	37	34	38	52	51	40	28	32
2144	41	22	26	35	49	35	42	23
2176	26	26	33	36	52	40	45	49
2208	50	34	41	35	44	25	23	23
2240	31	29	39	46	50	36	31	32
2272	42	32	34	41	44	33	43	30
2304	31	28	39	46	50	40	35	37
2336	43	35	33	43	43	29	37	29
2368	27	22	30	33	43	31	32	26
2400	36	31	31	29	40	28	26	22
2432	29	42	48	51	65	52	43	54
2464	63	34	42	44	56	33	38	32
2496	26	22	23	38	49	32	26	38
2528	43	22	24	27	45	24	26	17

Address	Value (+0)	Value (+4)	Value (+8)	Value (+12)	Value (+16)	Value (+20)	Value (+24)	Value (+28)
2560	36	36	47	51	58	58	43	42
2592	43	28	37	43	56	48	38	32
2624	28	19	28	34	46	34	28	34
2656	45	20	29	28	50	32	26	21
2688	37	38	51	50	55	45	38	49
2720	56	28	38	40	50	29	44	26
2752	32	35	50	43	53	44	41	41
2784	34	24	35	34	39	33	34	29
2816	21	33	31	45	48	42	27	29
2848	40	17	21	32	45	30	29	26
2880	26	27	38	33	29	31	32	31
2912	35	25	29	29	34	15	25	23
2944	34	28	44	45	41	41	37	45
2976	45	17	34	44	46	30	43	29
3008	31	36	37	50	54	44	28	40
3040	38	22	27	28	45	32	36	22

0.4.4 dataset50.h

Couldn't compile due to resource limitations. Skipped.

0.4.5 Diff validation

```
~/.../02_labs/l1 % P main U:11 7:10 % whirck@TPX13-YG3-WR 14:40:27
ls
asm_outputs assembly_practice datasets
~/.../02_labs/l1 % P main U:11 7:10 % datasets.zip 'lab 1 RISC-V Matrix-Matrix Multiplication Lab.pdf' matrix_multiply.asm report test.sh
% cat test.sh
#!/bin/bash

diff ./datasets/d3_soln.txt ./asm_outputs/output_TC3.txt
diff ./datasets/d10_soln.txt ./asm_outputs/output_TC10.txt
diff ./datasets/d16_soln.txt ./asm_outputs/output_TC16.txt
diff ./datasets/d50_soln.txt ./asm_outputs/output_TC50.txt

~/.../02_labs/l1 % P main U:11 7:10 % whirck@TPX13-YG3-WR 14:40:33
% ./test.sh
diff: ./asm_outputs/output_TC50.txt: No such file or directory
~/.../02_labs/l1 % P main U:11 7:10 % x 2 whirck@TPX13-YG3-WR 14:40:38
```

Figure 1: Shell validation using ‘diff’